ECO491/FIN591 – Problem Set 6: Option Full Valuation and Backtesting

Shayla Zheng

December 12, 2025

1 1 Currency Option Risk

1.1 1.1 Risk factor plots and HS description

Risk factors are (S, r_d, r_f, σ). Traditional HS creates daily shocks $(\Delta \ln S, \Delta r_d, \Delta r_f, \Delta\sigma)$ from the historical series, applies each shock to today's factors, re-prices the option for each scenario (full valuation), and takes $\mathrm{VaR}_p = -Q_p(\Delta c)$.

figs/p1_spot.png

Figure 1: USDBRL spot.

figs/p1_rates.png

Figure 2: Interest rates.

`figs/p1_iv.png`

Figure 3: Implied volatility (3M).

1.2 1.2 HS $\mathrm{VaR}_{1\%}$ (full window, full valuation)

For $S_0 = K = 3.919$ and $T = 0.25$, using the last-date $(r_{d,0}, r_{f,0}, \sigma_0)$ from the spreadsheet:

$$\boxed{\mathrm{VaR}_{1\%} \approx 0.056437, \quad c_0 \approx 0.214807.}$$

figs/p1_pnl_hist.png

Figure 4: HS P&L histogram (full valuation).

2 2 Backtesting Value at Risk (Apple)

2.1 2.1 VaR series and hit sequences

Define hits by $I_{t+1} = \mathbf{1}\{R_{t+1} < -\mathrm{VaR}_{t+1}\}$. Figures below show VaR and violations for HS (250d) and RiskMetrics EWMA ($\lambda = 0.94$).

figs/p2_hs_var.png

Figure 5: HS VaR and returns (violations marked).

figs/p2_hs_viol.png

Figure 6: HS violations only.

figs/p2_rm_var.png

Figure 7: RiskMetrics VaR and returns (violations marked).

figs/p2_rm_viol.png

Figure 8: RiskMetrics violations only.

2.2 2.2 UC test

Critical value at 10%: 2.7055. Results (computed in the accompanying code):

Method	T	T_1	$\hat{\pi}$	LR_{uc}	Reject at 10%?
HS (250d)	1259	18	0.0143	2.0724	No
RiskMetrics (EWMA)	1259	28	0.0222	14.1322	Yes